Case report

Hypotension: an unusual presentation of vitamin B_{12} deficiency, with complete recovery following cyanocobalamin therapy

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SUMMARY

Vitamin B_{12} deficiency and its sequelae are well described and reported, especially in vegetarians. However, its association with haemodynamic instability is not well identified. We report a case of a young man, previously healthy, presenting with fever, hypotension requiring vasopressors and pancytopenia. Extensive workup was unrevealing for possible infective, inflammatory or endocrine causes except for vitamin B_{12} deficiency. Fever and haematological parameters stabilised after adequate supplementation of cyanocobalamin (vitamin B_{12}).

BACKGROUND

Vitamin B_{12} is an essential micronutrient. Its deficiency is associated with anaemia, neurological disease and various forms of cancer.[1] Vegetarian diet, mainly vegan diet, has been associated with low serum B_{12} levels.[2] An association between vitamin B_{12} deficiency and autonomic neuropathy has been suggested.[3] Autonomic neuropathy can impair the body's capacity to adjust vascular resistance leading, in turn, to pooling of venous blood, a subsequent decrease in stroke volume and cardiac output.**4** Near syncope has a reported association with vitamin B_{12} deficiency.[3 5] Hypotension associated with delirium has previously been reported in an elderly with a vegetarian diet.[6] To the best of our knowledge, however, the association between vitamin B_{12} deficiency and hypotension has not been described in a young patient before.

We are reporting a case of megaloblastic anaemia, secondary to vitamin B_{12} deficiency, presenting with fever, pancytopenia, haemolysis and hypotension, with dramatic response to cyanocobalamin supplementation.

CASE PRESENTATION

A 27-year-old previously fit and well Nepalese man, who had been a strict vegetarian throughout his life, presented with a 4-week history of off/on low-grade fever and occasional vomiting. It was associated with gradually worsening yellow discolouration of the skin, eyes and urine along with pale stools. This presentation was preceded by a 3-month history of poor appetite, approximately 10 kg unintentional weight loss and tingling in his hands and feet.

On presentation, he appeared unwell and was found to be febrile, tachycardiac and hypotensive (blood pressure=79/40 mmHg, heart rate=140/min, respiratory rate=20/min, temperature=39.5ºC and O2 saturation=99% at room air). As part of the sepsis protocol, intravenous fluids and antibiotics were started. Examination was remarkable for pallor and scleral icterus. No lymphadenopathy or hepatosplenomegaly was appreciated, and neurological examination was unremarkable. Initial workup revealed severe megaloblastic anaemia, pancytopenia and coagulopathy with low serum B_{12} level (table 1). He was initiated on norepinephrine infusion as his hypotension was non-fluid responsive. Two units of packed red cells were transfused.

The patient remained febrile despite broad-spectrum antibiotics. On day 2 of hospitalisation, intramuscular cyanocobalamin 1000 µg per day, was initiated. His blood pressure improved following vitamin B_{12} supplementation and norepinephrine infusion was stopped (table 2).

INVESTIGATIONS

Complete workup for a source of fever including multiple sets of blood cultures, respiratory viral panel, two malarial blood films (thin and thick), dengue serology, viral hepatitis serology (A, B, C and E) and HIV screen were negative with a relatively low C-reactive protein of 10.9 mg/L and normal serum lactate. Quantiferon gold test for tuberculosis along with Epstein-Barr virus, adenovirus and cytomegalovirus-PCR of the serum were negative. Serum homocysteine and thyroid-stimulating hormone levels were within normal limits, and so were serum adrenocorticotropic hormone (ACTH) and cortisol levels. The ECG at presentation revealed sinus tachycardia otherwise unremarkable.

Workup for his anaemia revealed a marked macrocytic anaemia with anisocytosis, hypersegmented neutrophils, macro-ovalocytes and leucoerythroblastic picture on peripheral blood smear. The findings were associated with vitamin B_{12} deficiency (figure 1). Iron profile in the serum, including iron, ferritin, transferrin saturation and serum folate, was typical. His vitamin B_{12} level was below the detectable threshold of 73.8 pmol/L (normal level: 144–596 pmol/L). Anti-intrinsic factor antibodies were negative. His serum lactic dehydrogenase level was high (>1800 U/L) with elevated serum total bilirubin (mainly unconjugated) and normal serum transaminases, consistent with haemolysis. Direct Coombs test was positive; however, it followed the

Table 1 Relevant laboratory tests

	At admission	1 week of B_{12} therapy	2 weeks of B_{12} therapy	Reference range
White blood count (K/uL)	2.6	2	5.4	4.5–11
Red blood count ($Kx10^6$/ul)	0.9	2.9	3.5	4.3–5.9
Haemoglobin (gm/dL)	4	9.5	12	13.5–17.5
Mean corpuscular volume (fl)	122.8	99	103.4	80–100
Hct (%)	11.3	28.6	36.1	41–53
Mean corpuscular haemoglobin (pg)	43.5	32.9	34.4	25.4–34.6
Mean corpuscular haemoglobin concentration (g/dL)	35.4	33.2	33.2	31–36
RDW-CV (%)	24	25.2	19.9	11.5–14.5
ANC ($Kx10^3$/µL	1.6	0.2	2.7	1.5–8
Lymphocyte ($Kx10^3$/µL)	0.6	1.4	1.9	1–4.8
Monocyte ($Kx10^3$/µL)	0.3	0.3	0.6	0.2–1
Platelet (10^9/L)	91	82	728	150–400
Retic count ($Kx10^3$/µL)	9	50	106.4	50–100
Retic %	1.1%	1.8%	3.8%	0.5–1.5
INR	1.7	1.2	1.2	0.8–1.1
PT (s)	17.5	12.4	N/A	11–15
PTT (s)	42.8	28.7	N/A	25–40
Total bilirubin (µmol/L)	66	33	28	2–17
Direct bilirubin (µmol/L)	8.10	15.4	N/A	0–5.1
B_{12} level (pmol/L)	< 73.8	>1476	N/A	148–590
Folate (nmol/L)	17	N/A	N/A	5.7–45.3

ANC, absolute neutrophil count; HCT, hematocrit; INR, international normalized ratio; PT, prothrombin time; PTT, Partial thromboplastin time; RDW-CV, red cell distriction width-coefficient of variation.

blood transfusion. Elution test was carried out and was negative, indicating reactive positivity. Autoimmune and G6PD deficiency screen were negative. Flow cytometry of peripheral blood was negative for paroxysmal nocturnal haemoglobinuria.

Also, a contrast-enhanced CT scan of the chest, abdomen and pelvis was performed, which was unrevealing.

Bone marrow biopsy and aspiration after 1 week of intramuscular vitamin B_{12} supplementation was negative for tuberculosis and atypical leukaemic cells with a normal karyotype on chromosome analysis.

Table 2 Mean arterial pressure with time

Time in hours	Event	Mean arterial pressure
0	Presentation and initiation of norepinephrine	53
8	Still on norepinephrine	85
36	First dose of cyanocobalamin	83
38	Discontinuation of vasopressors	87
132	3 days after first dose of B_{12}	90
468	Outpatient follow-up (2 weeks)	96

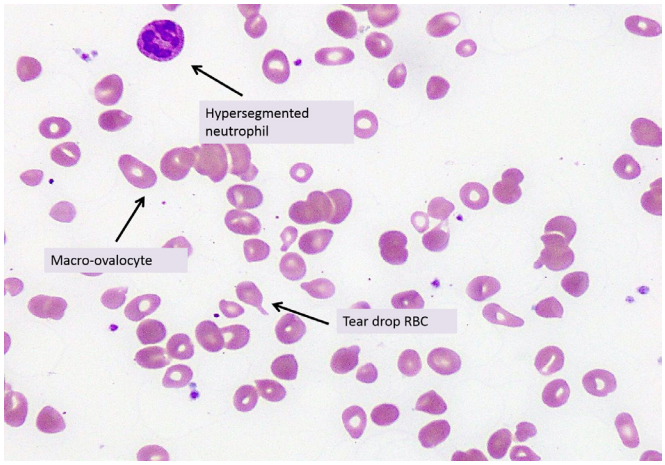


Figure 1 Peripheral blood smear shows marked megaloblastic anaemia with anisocytosis, teardrop cells, hypersegmented neutrophils and macro-ovalocytes, findings consistent with vitamin B_{12} deficiency.

Differential diagnosis

Out of the extensive differential list, sepsis was on top of the list initially due to fever and hypotension. It was ruled out based on multiple sets of cultures that turned out to be negative, low lactate and low inflammatory markers. There still remains a possibility of an underlying difficult-to-culture organism. It is less likely considering the patient's continued improvement despite discontinuation of antibiotics on day 2 of admission. This was advised by the infectious disease team after a review of the preliminary negative cultures and low inflammatory markers. In our opinion, the duration of antibiotics was not sufficient to treat an infection causing this degree of haemodynamic compromise. Acute viral illness, including dengue fever, was investigated by viral serology, including antibody detection for dengue virus. Adrenal insufficiency was a consideration due to unexplained hypotension and was ruled out by normal levels of serum cortisol and ACTH. Serum troponin, ECG and transthoracic echocardiogram were unremarkable. A cardiac cause for his haemodynamic instability was not evident.

With regards to haemolysis and pancytopenia, folate deficiency and paroxysmal nocturnal haemoglobinuria were ruled out by a normal serum folate level and negative flow cytometry of peripheral blood.

An underlying bone marrow disorder like leukaemia and lymphoma were excluded after bone marrow biopsy, flow cytometry of peripheral blood and CT scan of the chest, abdomen and pelvis.

An extensive, yet unrevealing workup led to the final working diagnosis of B_{12} deficiency. The supporting evidence was there in the form of significant improvement in haemodynamic parameters, symptoms, haematological parameters and overall picture after receiving vitamin B_{12} supplementation.

TREATMENT

The patient was initiated on intramuscular supplementation of cyanocobalamin 1000 mcg once daily. Later he switched to oral tablets.

OUTCOME AND FOLLOW-UP

The patient's fever, pancytopenia, coagulopathy and blood pressure recovered entirely after 2 weeks of vitamin B_{12} replacement. His appetite improved, and he started to gain back some of his lost body weight.

DISCUSSION

Vitamin B_{12} deficiency has been reported to be associated with reversible autonomic disturbance. Sympathetic activation is responsible for the autonomic disturbance.[7] Orthostatic hypotension in patients with vitamin B_{12} deficiency has been reported even in the absence of neurological and haematological manifestations.[7 8] Vestibulosympathetic reflex leading to increased sympathetic activity and vagal activity decrease are the primary mechanism by which the body increases peripheral resistance. This increase in peripheral resistance leads to increased venous return, and in turn, increased cardiac output, leading to increased blood pressure. We propose a mechanism consistent with inadequate autonomic response caused by vitamin B_{12} deficiency, leading to hypotension with the lack of appropriate autonomic response by the body. In our case, the response of the patient's blood pressure to volume expansion is consistent with our proposed mechanism physiologically.[4]

Homocysteinemia is a modifiable risk factor for cardiovascular disease. It damages the vascular endothelial cells, stimulates vascular smooth muscle proliferation, enhances low-density lipoprotein cholesterol peroxidation and thrombosis formation.[9 10] Vegetarians have been reported to have an elevated homocysteine in the context of B_{12} deficiency.[11 12]

The association between vitamin B_{12} deficiency and fever has been described previously.[13] Fever, defined as a temperature of 100°F (37.8°C) or more, was present in approximately 40% of the patients with megaloblastic anaemia secondary to vitamin B_{12} deficiency. Furthermore, a temperature of over 104°F (40°C) was reported, in subjects with severe megaloblastic anaemia, thrombocytopenia and elevated bilirubin concentration.[14]

Patient's perspective

I followed a healthy lifestyle and was never really sick before. The strict vegetarian diet is due to my religious beliefs. The medical team advised cereals fortified with vitamin B_{12}; however, I did not like the taste. The dietician advised dried purple laver, but it is not available in Qatar. I wish there were enough public awareness in vegetarians about this condition so we could get tested and treated in time.

Learning points

- Strict vegetarian diet can lead to severe vitamin B_{12} deficiency with levels below detectable limit by available assay even if the dairy intake is adequate.
- Vitamin B_{12} deficiency can present not only with anaemia, but also with pancytopenia and otherwise unexplained fever and neuropathy.
- Autonomic dysfunction can be a manifestation of B_{12} deficiency.
- B_{12} replacement is relatively easy. We can prevent the deficiency with timely screening in the at-risk population.

B_{12} is readily available in dairy source. A glass of milk can provide up to 50% of recommended daily allowance.[15] Our patient had a regular intake of dairy products including cow milk and yoghurt two to three times a week. He developed severe B_{12} deficiency despite the intake of milk and yoghurt. Authors' are of the opinion that the amount of B_{12} in milk varies with the genotype of the cow producing the milk.[16]

We propose that the fever, pancytopenia and autonomic dysfunction manifesting as hypotension were all consequences of vitamin B_{12} deficiency in our patient, secondary to a strict vegetarian diet. The dramatic improvement following vitamin B_{12} replacement is in keeping with our hypothesis. We believe that regular screening for vitamin B_{12} deficiency in patients with known risk factors, such as strict long-term vegetarianism, can help in detecting its deficiency in the early stages and in providing adequate treatment to prevent these severe consequences.

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